

CHAPTER 1

Chemical Reactions and Equations



1064CH01

Consider the following situations of daily life and think what happens when –

- milk is left at room temperature during summers.
- an iron tawa/pan/nail is left exposed to humid atmosphere.
- grapes get fermented.
- food is cooked.
- food gets digested in our body.
- we respire.

In all the above situations, the nature and the identity of the initial substance have somewhat changed. We have already learnt about physical and chemical changes of matter in our previous classes. Whenever a chemical change occurs, we can say that a chemical reaction has taken place.

You may perhaps be wondering as to what is actually meant by a chemical reaction. How do we come to know that a chemical reaction has taken place? Let us perform some activities to find the answer to these questions.

Activity 1.1

CAUTION: This Activity needs the teacher's assistance. It would be better if students wear suitable eyeglasses.

- Clean a magnesium ribbon about 3-4 cm long by rubbing it with sandpaper.
- Hold it with a pair of tongs. Burn it using a spirit lamp or burner and collect the ash so formed in a watch-glass as shown in Fig. 1.1. Burn the magnesium ribbon keeping it away as far as possible from your eyes.
- What do you observe?

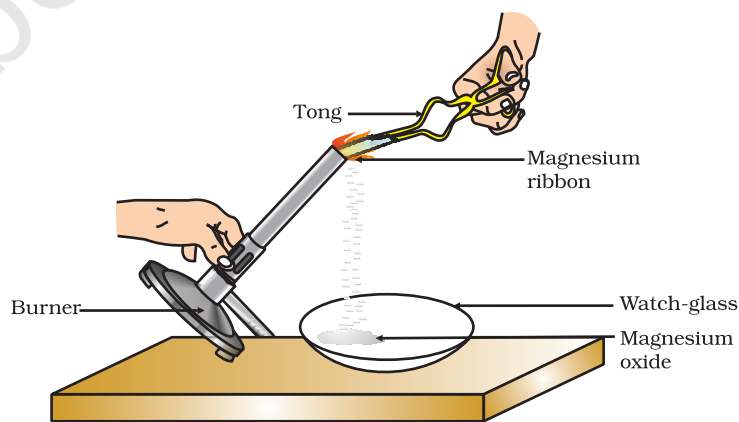


Figure 1.1

Burning of a magnesium ribbon in air and collection of magnesium oxide in a watch-glass

Table: user_information Use alias: ui			
id	user_id	full_name	email
1	10111	Tinker Bell	tb@xyz.com
2	10112	Jasmine T	jt@abc.com
3	10113	Rosetta PW	rpw@abc.com
4	10114	Zarina Leo	zl@xyz.com
5	10115	Peter Cyrus	pc11@abc.com
6	10116	Pan Peter	pp@xyz.com
7	10117	Daisy T	dt_dt@xyz.com
8	10118	Captain Hook	capt@xyz.com
9	10119	Marina	marina@abc.com

Table: user_complaints Use alias: uc				
complaint_id	user_id	category_id	subcategory_id	created
1	10114	5	5c	2023-05-20
2	10115	3	3b	2023-05-09
3	10117	1	1b	2023-05-17
4	10115	3	3a	2023-03-05
5	10111	4	4b	2023-01-26
6	10120	1	1d	2023-01-26
7	10116	4	4a	2023-04-08
8	10117	2	2a	2023-01-03
9	10112	4	4b	2023-03-31
10	10115	5	5a	2023-05-11

Table: complaint_category Use alias: cc	
category_id	category_name
1	driver behavior
2	overcharging
3	poor navigation
4	cleanliness
5	payment issues

Table: category_description Use alias: cd			
id	category_id	subcategory_id	reason
1	1	1a	unprofessional
2	1	1b	drunk
3	1	1c	reckless driving
4	1	1d	rude
5	2	2a	taking longer routes
6	2	2b	tampering with fares
7	2	2c	demanding toll charges
8	3	3a	taking incorrect routes
9	3	3b	relying too much on GPS
10	3	3c	drivers getting lost
11	4	4a	unclean vehicles
12	4	4b	unpleasant odors
13	5	5a	disputed charges
14	5	5b	payment failures
15	5	5c	issues with refunds

Table: complaint_status Use alias: cs			
id	complaint_id	status	updated
1	5	in review	2023-01-26
2	5	resolved	2023-01-27
3	10	in review	2023-05-12
4	7	in review	2023-04-08
5	7	in progress	2023-05-18
6	9	in review	2023-03-31
7	9	failed to resolve	2023-04-02
8	3	in review	2023-05-17
9	3	resolved	2023-05-17
10	8	escalated	2023-05-19
11	2	in review	2023-05-09
12	2	in progress	2023-05-09
13	1	in review	2023-05-20
14	1	escalated	2023-05-20
15	6	in review	2023-01-27
16	6	failed to resolve	2023-02-15
17	4	contacted user	2023-03-06
18	4	legal action taken	2023-03-10
19	15	in review	2023-05-08
20	17	in review	2023-02-01

Table: user_rides Use alias: ur								
id	user_id	start_time	end_time	distance	fare	ride_status	date	rating
1	10112	08:00	08:30	3	150	ongoing	2023-05-30	NA
2	10114	10:30	10:45	1	50	completed	2023-05-11	95
3	10115	22:30	23:10	5	250	ongoing	2023-05-30	NA
4	10118	19:15	19:50	6	300	completed	2023-05-10	100
5	10114	08:00	09:00	10	500	completed	2023-05-20	0

Table: driver_feedback Use alias: df					
fd_id	driver_id	user_id	rating	comment	date
1	151	10112	30	asking personal questions	2023-05-30
2	172	10114	95	fun ride	2023-05-11
3	158	10115	10	took different route	2023-05-30
4	194	10118	100	awesome driving	2023-05-10
5	172	10114	0	trying to get too friendly	2023-05-20

Table: customer_segments Use Alias: csg	
segment_id	segment_name
1	Business
2	Tourist
3	Students
4	Families
5	Solo Travelers

Table: drivers Use alias: dr					
id	name	city_id	avg_rating	experience_yrs	current_status
151	Paul	1	1000	10	active
172	Sam	2	1500	8	on leave
158	Ava	3	2200	3	active
194	Cleo	3	200	4	inactive
156	Skipper	2	150	1	active

Table: cities Use Alias: ct	
city_id	city_name
1	New York
2	London
3	Tokyo
4	Mumbai
5	Sydney

Table: partner_companies Use Alias: pc		
company_id	company_name	industry
1	Hotel Chain A	Hospitality
2	Car Rental B	Transportation
3	Food Delivery C	Food & Beverage
4	Tour Package D	Travel
5	Retail Brand E	Retail